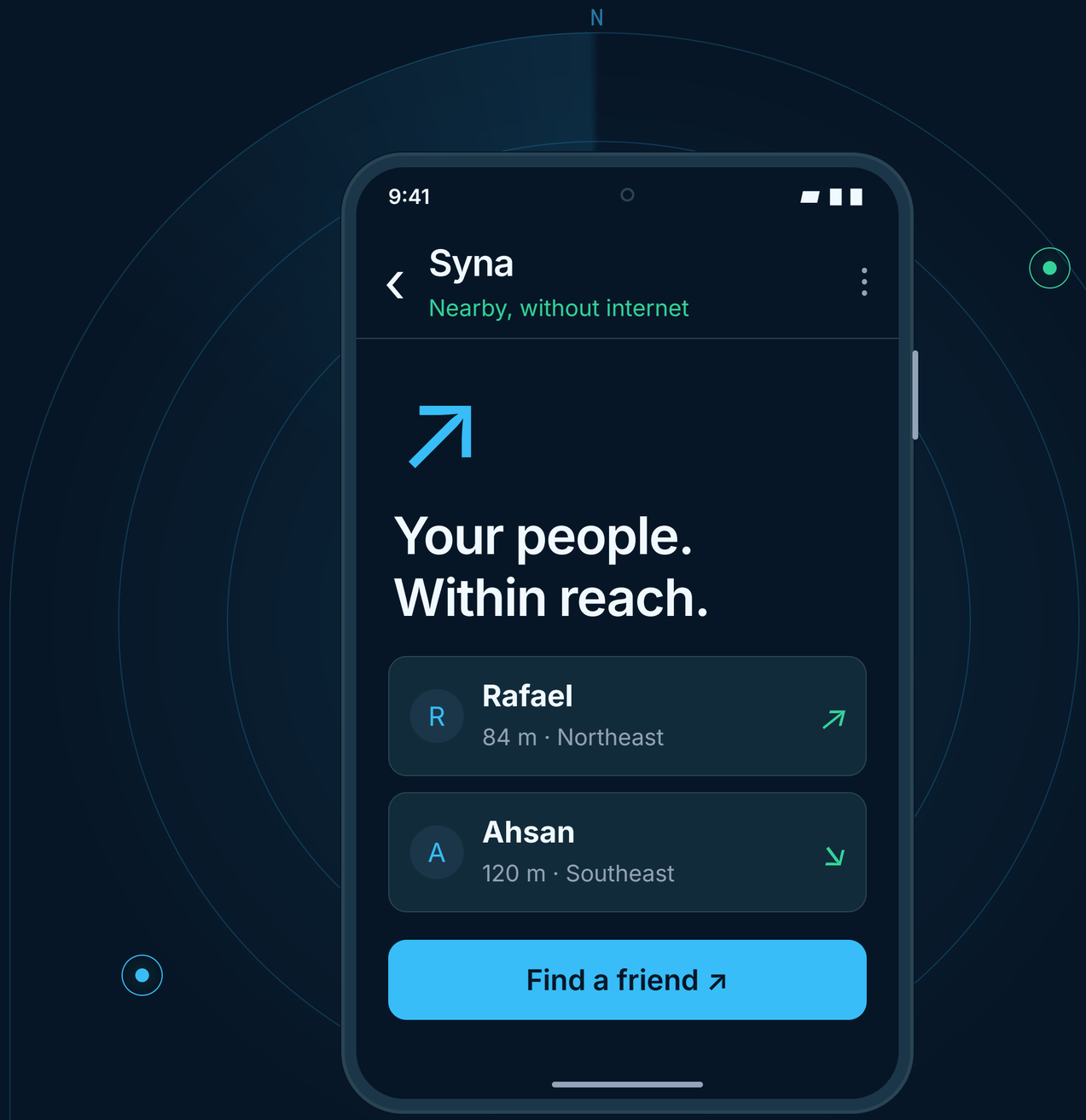
syna

Stay close. Even Offline.

Secure Offline P2P Messaging & Friend Finder

By LGTM Team



syna

Team Introduction



Rafael

Networking & Security

- Nearby Connections API integration
- P2P discovery
- Device handshake
- Public-key encryption
- Background GPS sensor streams

Experience: [add yours]



Andrew

Backend & Business Logic

- Room schema architecture
- Profile & contact CRUD
- Message persistence
- QR key management
- Distance and bearing mathematical logic

Experience: [add yours]



Ahsan

Frontend & UI/UX

- Jetpack Compose screens
- QR scanner UI
- Chat interface
- Offline MapLibre integration
- Animated compass radar UI

Experience: [add yours]

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Motivation & pain points

No signal.
Where is everyone?



Zero internet coverage.

Hiking, festivals, or natural disasters sever standard cellular connectivity.



Location sharing stops working.

No internet, no location updates.



Friends are hard to find.

No intuitive visual directional guidance.

syna

The solution

Syna: Off-grid connectivity.



Offline P2P Chat

Encrypted direct messages between nearby Android devices



Local Map Logs

Offline maps with locally saved, last-known friend locations



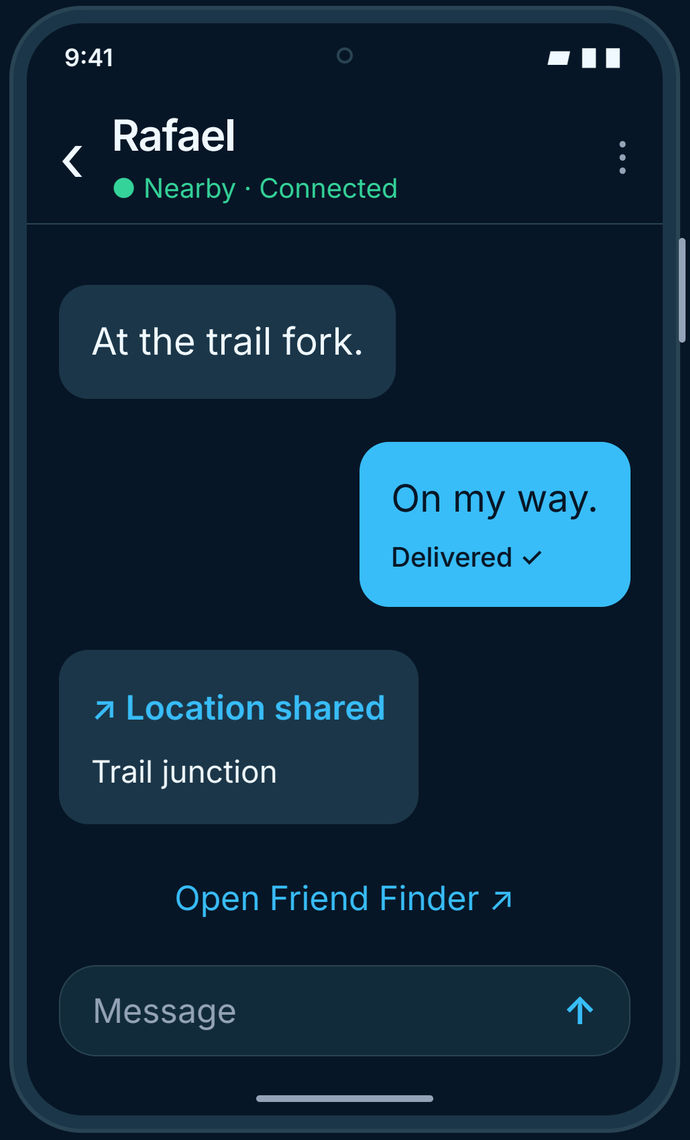
Radar Compass View

360° Friend Finder with directional edge glow

syna

Offline?
Keep the conversation going.

- Direct to your friend.
- Encrypted in transit.
- Saved on your phone.

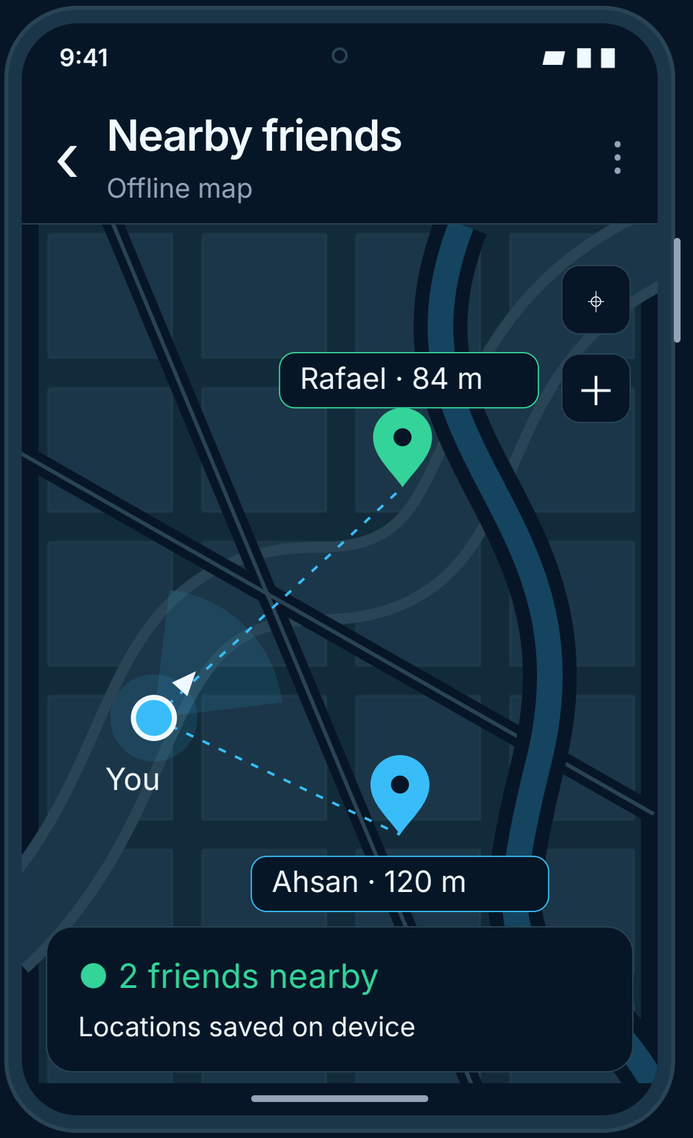


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See who’s nearby.

Nearby Connections API

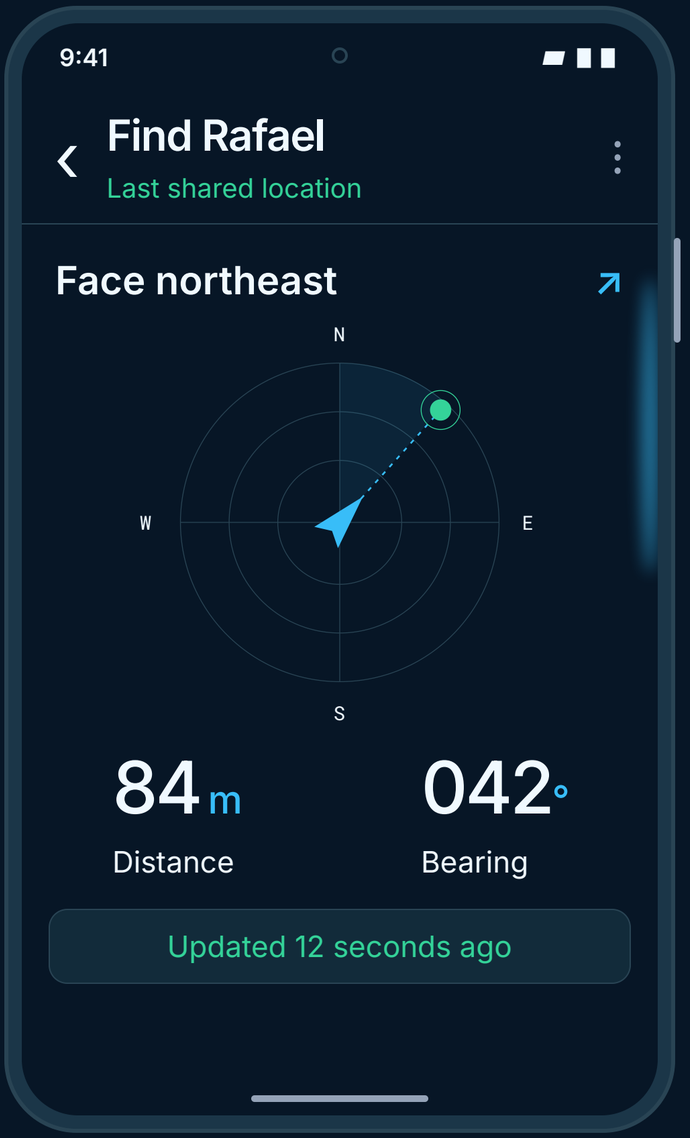
- Friends + distance.
- Your position + heading.
- Maps without internet.



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Find your friends. Know which way.

- Turn toward the glow.
- Read distance + bearing.
- Check location freshness.



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More than an offline message.

Feature Dimension	Messaging-first P2P Apps	Syna P2P Radar & Chat
Primary Focus	Text broadcasting & crisis messages	Offline P2P chat + directional friend finding
UI Experience	Standard 2D chat list view	Dynamic 360° Compass Radar View
Sensor Scope	Bluetooth / Wi-Fi radios	GPS Hardware + Magnetometer
Data CRUD Scope	Store & forward messages	Local chat storage + last-known coordinates

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Two devices.
One direct connection.

Your Android

Jetpack Compose + Material 3
Interface

Kotlin · MVVM
Application logic

Room (SQLite) + Keystore
Local data · Private keys

Nearby Connections
(Google API)

Encrypted P2P



Friend’s Android

Same local stack

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Tech stack & resources

Requirements

P2P transport

Nearby Connections API

Wi-Fi · Bluetooth · BLE

Security & QR

Android Keystore

ECDH key exchange (planned)
ZXing / ML Kit QR scanner

Local database

Room (SQLite)

Profiles · chats · contacts · waypoints

Offline mapping

MapLibre SDK

Prepackaged offline tiles

Sensors & radar

GPS · Android Location Services

Rotation Vector Sensor
Compose Canvas · Ambient edge glow

Build & test

Android Studio · Git / GitHub

Two physical Android phones

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Project Plan

Build first

Identity + contacts

Encrypted 1:1 chat

Offline map + compass

Beyond this MVP

Mesh relaying

Groups + file sharing

Calls + cloud sync

Nearby range required. Last locations stay saved.

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4-Week Implementation Plan

WEEK 1 · Nov 4–10

Foundation

- Project architecture
- Room schema
- Wireframes
- Raw offline P2P payload exchange test

WEEK 2 · Nov 11–17

Messaging

- QR identity exchange
- Authenticated encrypted chat
- Local Room persistence
- Contact CRUD

WEEK 3 · Nov 18–24

Location

- GPS sharing stream
- MapLibre tile loading
- Distance/bearing math
- Compass Radar view

WEEK 4 · Nov 25–Dec 1

Integration

- End-to-end testing
- Error handling
- Permission edge cases
- Live demo stabilization

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Thank You

Stay connected. Find each other. Even offline.

LGTM Team

